

"Good" XAI Design: For What? In Which Ways?

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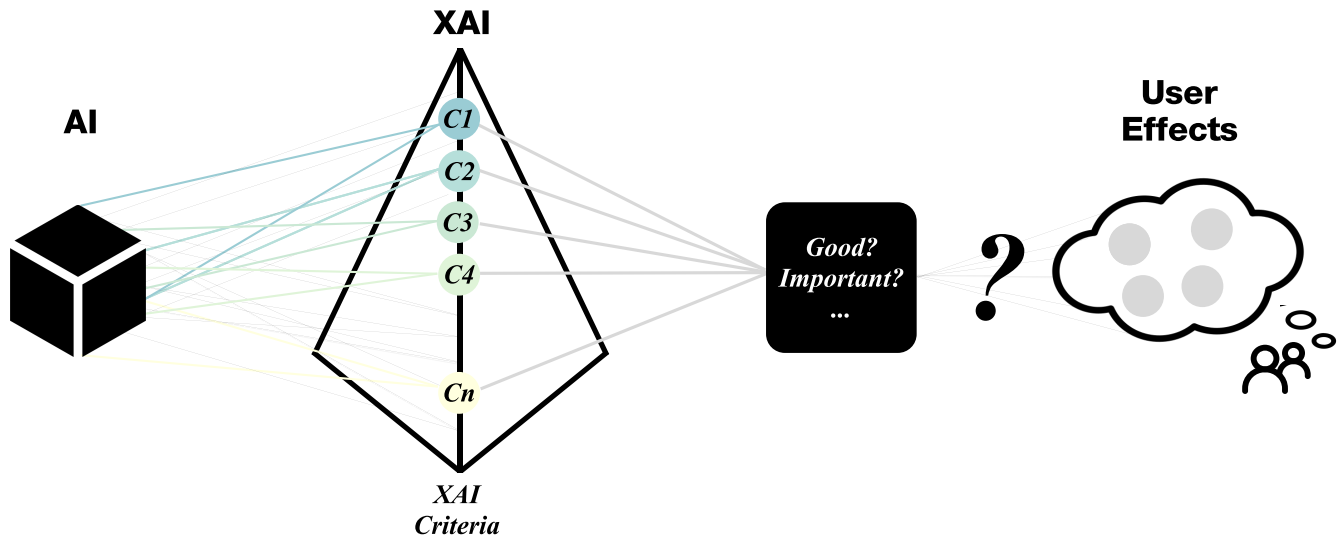


Figure 1: This figure illustrates the relationship among AI, XAI, its design criteria, and user effects. Acting as a prism, XAI transforms the often “black box” AI into explanations that humans receive. This transformation is typically guided and evaluated by a set of criteria (C1, C2, C3, etc.) which are deemed “good” and “important,” with little understanding of the relationship between these recommended XAI designs and their effects on users.

Abstract

The rapid proliferation of Explainable Artificial Intelligence (XAI) has led to the development of numerous evaluation frameworks aimed at guiding and optimizing its design. However, these frameworks often emphasize the technical properties of XAI artifacts, overlooking the nuanced perceptions and values of end-users. Recognizing that XAI impacts society and individuals in non-neutral ways, this study adopts a human-centered approach to systematically examine the effects of recommended XAI properties on the general public in everyday scenarios through a formative study involving 87 end-users. The findings reveal that comprehensibility is the most valued XAI property, while frequently advocated properties like contrastivity may have overall negative effects. These results highlight the necessity of a goal-driven reverse engineering approach that integrates human values into XAI design to ensure positive user outcomes. This paper bridges the gap between XAI

design and its impact on end-users, offering practical guidance for user-centered XAI development in everyday contexts.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **User centered design**; *Empirical studies in interaction design*; • **Computing methodologies** → Artificial intelligence.

Keywords

Explainable AI, XAI, user-centered design

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1 Introduction

Artificial Intelligence (AI) has been extensively integrated into various facets of daily life, exerting transformative effects on individuals and society at large. In response to the increasing complexity and pervasive influence of AI systems, researchers and practitioners

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have proposed Explainable AI (XAI) as a solution. Currently, numerous evaluation criteria frameworks exist, aiming to establish a common foundation and direction for guiding and optimizing XAI design [16, 40, 43, 52]. However, while these frameworks purport to support the development of "good" XAI, they predominantly focus on the properties of the artifacts themselves, with assumptions about what constitutes "good" for human users remaining largely vague or absent.

XAI is fundamentally a socio-technical system that impacts a wide range of stakeholders [18, 19, 21], mediating user interactions in non-neutral ways by prioritizing, reinforcing, and suppressing certain values intentionally or unintentionally [13, 20]. Taking a human-centered view, we argue that it is urgent to examine the impacts of the currently advocated XAI evaluation and design criteria—especially those that guide technical development—on users. This examination is crucial for understanding the impacts these systems might implicitly support or counteract. Considering the contingency of XAI design—that artifacts do not operate in isolation but are influenced by users [44], application contexts [40], and other factors [49, 56, 61]—we focus on everyday scenarios for end-users. This perspective has garnered increased interest recently (e.g., [5, 32, 38]), moving beyond the traditional focus on domain experts and high-stakes scenarios. To systematically explore the user effects of XAI properties advocated in the literature, we conducted a formative study with 87 end-users recruited from a crowd-sourced platform. Participants were asked to evaluate a set of XAI properties, rating each based on seven perceived effects and the overall importance. The results indicate that comprehensibility and coherence are the most fundamental and important properties of XAI, whereas anthropomorphism is the least favored. In addition, contrastivity, frequently advocated by researchers, may not be applicable to daily scenarios. Conversely, we advocate for goal-driven reverse engineering that incorporates human values and benefits users in the development of XAI. This approach suggests that, to achieve desired user effects and avoid adverse or unethical outcomes, a weighted combination of XAI properties should be prioritized in technical development.

This study makes the following contributions:

- Gap alignment: Connecting the techno-centric requirements of XAI artifacts with the human-centered impacts on end-users;
- User-centered XAI in everyday contexts: Providing empirical evidence on how XAI properties affect users in daily scenarios;
- Goal-oriented XAI engineering: Offering practical guidance on prioritizing criteria for XAI systems designed to support specific human goals.

2 Literature Review

2.1 Criteria for "Good" XAI

The evaluation standards of a field set the direction for its advancement and shape its future development. To establish a shared goal of validating and benchmarking XAI methods, the XAI field has actively discussed, developed, and debated evaluation constructs, criteria, or desiderata for XAI [16, 40, 43, 52]. For instance, based on

a systematic literature review, Nauta et al. [43] identified 12 properties of XAI that constitute a "good" explanation. They argue that such a conceptual framework enables objective and quantifiable evaluation, thereby fostering the robust advancement of the field, in contrast to approaches that rely on anecdotal evidence.

However, the assumptions underlying why these evaluation criteria are considered "good" [42] or "important" [40] are often implicit, vague, or even unstated, frequently relying on researchers' intuition [42]. For human-centered aspects, there is a lack of systematic investigation into why these computationally abstract criteria improve artifacts—whether they truly benefit humans, and if so, in what ways. This resonates with Fallman [23]'s argument that implicit notions of "good" and "bad" in design raise ethical concerns, as artifacts have non-neutral mediating effects on users. Similarly, Zhu et al. [63] argues that algorithm creation should incorporate stakeholders' values, goals, knowledge, and insights to reduce potential biases in design choices and prevent value compromises.

Previous work has frequently associated the "good" XAI as successfully explaining to users how the system works [5, 46], increasing users' trust [45, 54], and effectively decision making [10, 41]. Arrieta et al. [2] noticed scarce works have attempted to define XAI's goals from a conceptual perspective. They argue that XAI should target different goals depending on the intended audience. For users affected by model decisions, an important goal of XAI is to alleviate the burden experienced by non-technical or non-expert users when interacting with algorithms. Conversely, when the target users are domain experts, the primary goal of XAI shifts to facilitating the reuse of this knowledge in addressing other problems.

Building on these works, this study examines how XAI properties, recommended in the literature for design and evaluation, exert effects on end-users across everyday AI applications.

2.2 XAI in Everyday Scenarios

Recent research emphasizes the importance of integrating XAI into everyday AI applications. Studies show that XAI significantly influences how users incorporate AI into their daily routines [15]. In AR applications, which are becoming increasingly ubiquitous, Xu et al. [61] developed a framework that aligns XAI formats with AI output modalities. Kim [32] revealed that users leverage XAI explanations not only for understanding AI outputs but also for trust calibration, skill improvement, input optimization, and developer feedback. To enhance stakeholder engagement in XAI design, researchers have developed various design approaches. Lee et al. [38] created a card-based method introducing key explanation dimensions and facilitating co-creation with healthcare, finance, and management professionals. Similarly, Weitz et al. [60] employed co-creative workshops to identify XAI requirements for public sector AI users.

However, expanding the mainstream XAI design to everyday scenarios presents challenges due to context-dependent effectiveness. Many XAI research studies, especially those based on high-stake scenarios, typically focus on systematic reasoning and decision-making, aiming to minimize bias and optimize human-AI performance [59]. This has led to cognition-based designs that employ

forcing mechanisms to suppress bias, such as by extending the waiting time for explanations [10, 47]. While these approaches improve performance, studies indicate they can decrease user trust and increase cognitive load [10], limiting their applicability to everyday scenarios. It remains uncertain whether design principles established in prior studies—often focused on professional contexts—can be effectively adapted to more general, everyday environments [36].

Following calls for contextual XAI evaluation [40], our study examines end-users' perspectives on XAI properties across six common AI-integrated scenarios.

3 Methods

We conducted an IRB-approved online study via an online crowdsourcing platform Prolific¹ to collect end-users' views on embedding XAI into their daily scenarios. Following Liao et al. [40]'s work, participants (N=87, see demographics report in Appendix B.1) were first introduced to the definitions of design properties through training tasks in case they were unfamiliar with XAI. Then, they were asked to rate the effects of XAI design properties across all scenarios. Specifically, we selected six scenarios (see Appendix A.1 for details) in order to gain a more comprehensive view of everyday AI applications. In addition to the importance ratings used in Liao et al. [40]'s study, we further explored users' perceptions to illustrate XAI design's effects on users.

3.1 XAI Property and User Perception

To distinguish between the desired features of XAI artifact design and the effects of XAI on users, we use "XAI property" to refer to the recommended design and evaluation criteria for XAI artifacts, and "user perception" to represent the effects that XAI has on users (see Fig. 1 as well). The set of 9 XAI properties (i.e., comprehensibility, coherence, soundness, completeness, translucence, confidence, compactness, contrastivity, and anthropomorphism) was adapted from previous literature (e.g. [16, 40, 43]), with details shown in Appendix A.2, Table. 4.

To assess user perceptions, we used perceived importance as a measure to capture users' overall preference for each XAI property. Additionally, we adapted a set of seven user-centered effects (e.g., trustfulness and persuasiveness) [55] to further evaluate the impact of XAI design on end users. A total of 8 perceptions are measured using a 5-point Likert scale that ranges from -2 (indicating a negative impact) to 2 (indicating a positive impact), with 0 representing neutral effects. Although this set of selected effects may not be comprehensive, we believe it covers a broad range and is supported by existing literature that discusses these properties (Table 1).

In the within-subjects experiment, participants were asked to evaluate the effects of XAI properties (independent variables) on user perceptions (dependent variables) across six scenarios (independent variables).

3.2 Statistical Analysis

To investigate the relationship between the dependent variable (user perception) and the independent variable (XAI property and daily scenario), a group of generalized linear mixed effects models

was fitted for each perception variable. These models accounted for repeated measurements within individuals by including ProlificID as a random intercept, thereby controlling for unobserved heterogeneity across individuals. This approach ensured that the fixed effects of property and scenario were estimated while also considering within-individual correlations. The fixed effects for property (with "Comprehensibility" as the reference category) capture the effects of these property variables on perception values, while the random intercept accounts for variations specific to individual participants. The models were implemented using the *glmer* function from the *lme4* package in R [4].

4 Results

4.1 Tests for sample size and model choice

Our analysis started with verifying whether the sample size was adequate for meaningful inferences. Each model, focusing on one user perception, has 4,698 observations derived from 87 participants, with each participant rating 9 different XAI properties in 6 scenarios. Intraclass correlation coefficients (ICC) and R^2 values (see detailed results in Appendix B) validate the dataset's adequacy for detecting meaningful effects. Using 500 simulations, the post hoc power achieves 100% with confidence intervals between 96.38% and 100%.

Before applying the linear mixed-effects model, several assumption tests were conducted [26]. We first plotted the residuals against the explanatory variables to ensure a linear relationship with the response. An ordered residual plot was also examined, confirming no visual trend about the dependency of errors. The Breusch-Pagan test was performed and suggested that heteroskedasticity could not be ruled out, while the Kolmogorov-Smirnov test indicated deviations from normality.

Given the potential violations of normality and homoskedasticity, we decided to adopt a generalized linear mixed model (GLMM), with bootstrapping to ensure robust estimation and inference [9]. The bootstrapping procedure was implemented by repeatedly resampling the dataset with replacement over 1,000 iterations to capture variability in the data. Within each bootstrap iteration, GLMM was fitted to the resampled data, and the resulting coefficient estimates were collected to construct empirical distributions, from which 95% confidence intervals were derived using the 2.5th and 97.5th percentiles.

In each GLMM, we modeled one user perception as the dependent variable, incorporating fixed effects for XAI properties and scenarios, and adding a random intercept for participants to capture individual variation, resulting in eight models in total. Our chosen models outperformed the models without random effects, achieving a lower Akaike Information Criterion (AIC) and demonstrating enhanced fit.

4.2 Effects of XAI design properties on users' perceptions

In the section below, we present the results of running GLMM for each perception (dependent variable) on each property and scenario (independent variables), with 1,000 bootstrap iterations for the robustness of standard errors (e.g., Fig. 3). The mean ratings

¹<https://www.prolific.co/>

Effects	Definition	Example Literature
Scrutable	Allow users to tell the system it is wrong	[57]
Transparent	Explain to users how the system works	[5, 46]
Trustful	Increase users' trust in the system	[51, 54]
Persuasive	Convince users to do or believe something	[11, 17]
Effective	Help users make good decisions	[10, 41]
Efficient	Help users make decisions faster	[10]
Satisfying	Increase the ease of use or enjoyment	[8, 28]

Table 1: User Effects and their definition adapted from Tintarev and Masthoff [55], along with XAI literature that have discusses each effect.

of each perception by user participants for each property are also presented in Appendix B.2, Table 8.

Scrutability is most negatively affected by the anthropomorphism and compactness, suggesting that compact explanations relying on human-like elements could considerably hinder users' ability to judge the system performance. Specifically, the presence of anthropomorphism decreases the level of scrutability by 0.272 ($SE = 0.059, p < .001$) compared to the baseline (i.e., 'Comprehensibility'). The presence of compactness is associated with a reduction in the level of scrutability by 0.238 ($SE = 0.059, p < .001$), compared to the baseline. The effects of all other properties are not significantly different from the baseline.

Transparency is the perception most negatively affected by the anthropomorphic feature across all measured perceptions. Besides, among all properties, it is most strongly reduced by anthropomorphic explanations ($\beta = -0.337, SE = 0.051, p < .001$), suggesting that incorporating human-like elements in explanations can make people feel the least transparent. Transparency is also negatively associated with compactness ($\beta = -0.286, SE = 0.051, p < .001$) and confidence ($\beta = -0.119, SE = 0.051, p = .020$), indicating that condensed and confident explanations might obscure critical details, especially about limitations, and reduce perceived transparency. In contrast, translucence has the strongest positive effect on transparency ($\beta = 0.136, SE = 0.051, p = .008$). This is not surprising since the translucent property is designed to bring more perceived transparency to users.

Trustfulness can be negatively predicted by compactness ($\beta = -0.227, SE = 0.048, p < .001$) and anthropomorphism ($\beta = -0.219, SE = 0.048, p < .001$). This reveals that users are less likely to trust the explanations if they rely heavily on human-like elements and use overly concise language. The effects of all other properties on a trustful perception are statistically not different from the baseline.

Persuasiveness is significantly impacted by the compact ($\beta = -0.193, SE = 0.049, p = .015$) and anthropomorphic ($\beta = -0.121, SE = 0.049, p < .001$) properties, which are the top negative contributors. This implies that overly concise and heavily human-like explanations are less likely to convince users.

We find that confidence, translucence, contrastivity, anthropomorphism, and compactness negatively contribute to perceived **effectiveness** significantly. These XAI features should be avoided to deliver an effective user experience. For instance, compactness is negatively correlated with perceived effectiveness ($\beta = -0.141, SE = 0.045, p = .002$).

Likewise, confidence, translucence, contrastivity, anthropomorphism, and completeness negatively contribute to perceived **efficiency** significantly. These XAI features should be avoided to deliver an efficient user experience. For example, completeness is negatively correlated with perceived efficiency ($\beta = -0.134, SE = 0.043, p = .002$).

As for users' **satisfaction**, contrastivity ($\beta = -0.208, SE = 0.059, p < .001$), translucence ($\beta = -0.178, SE = 0.056, p = .001$), compactness ($\beta = -0.148, SE = 0.056, p = .008$), anthropomorphism ($\beta = -0.144, SE = 0.056, p = .010$), and confidence ($\beta = -0.127, SE = 0.056, p = .002$) all contribute negatively compared to the baseline property, "Comprehensibility."

Similarly, for users perceived **importance** of the explanation properties, all except coherence show statistically significant negative effects relative to "Comprehensibility." The most negative factors are anthropomorphism ($\beta = -0.500, SE = 0.052, p < .001$) and contrastivity ($\beta = -0.340, SE = 0.052, p < .001$).

5 Discussion

5.1 Reflections on Design Properties

In this session, we identify patterns in how design properties influence user perceptions. **Comprehensibility**, serving as the benchmark, emerges as the most desired and fundamental property, given most of the significant effects in Table 1 are negative. Meanwhile, **coherence** shows no statistical effect on any perception, suggesting that it holds a similar level of impacts to **comprehensibility**. Both properties are intrinsically linked to user understanding, echoing the emphasis on comprehension found in previous studies [7, 41, 45, 54]. While there's urgent need for increased public AI literacy [5], XAI may also reduce interpretability barriers by providing explanations that are accessible to a wide range of users.

Anthropomorphism is consistently associated with all perceptions, implying that human-like attributes detract from desired outcomes. Although previous research has argued that human-like traits promote heuristics that enhance human-AI communication [14, 39], and specifically in XAI, end-users prefer explanations that resemble human reasoning [33], our findings suggest otherwise. We propose two potential explanations for these results. First, AI that impersonates humans may discomfort and agitate users, especially as AI already stimulates concerns such as replicating the human workforce and manipulating online opinions [25, 29]. Second, our study design might have influenced the results; unlike the more

Perception	Scrutable	Transparent	Trustful	Persuasive	Effective	Efficient	Satisfying	Important
Property								
<i>Anthropomorphism</i>	-0.272***	-0.337***	-0.219***	-0.121*	-0.208***	-0.205***	-0.145**	-0.500***
<i>Coherence</i>	0.065	-0.003	0.036	0.000	-0.029	-0.046	-0.046	-0.035
<i>Compactness</i>	-0.238***	-0.286***	-0.227***	-0.193***	-0.141**	0.002	-0.149**	-0.284***
<i>Completeness</i>	0.057	0.049	0.019	-0.034	-0.063	-0.134**	-0.098	-0.117*
<i>Confidence</i>	-0.088	-0.119*	-0.029	0.009	-0.094*	-0.132**	-0.128*	-0.147**
<i>Contrastivity</i>	-0.027	-0.083	-0.086	-0.071	-0.157***	-0.193***	-0.208***	-0.340***
<i>Soundness</i>	0.057	0.080	0.055	-0.027	-0.066	-0.091*	-0.105	-0.108*
<i>Translucence</i>	0.070	0.136**	0.035	-0.077	-0.118**	-0.138**	-0.178**	-0.144**
Scenario								
<i>Finance</i>	0.034	0.029	0.038	0.033	0.029	0.021	-0.043	0.082
<i>Healthcare</i>	0.060	0.044	0.046	0.032	0.036	0.019	-0.060	0.111**
<i>Learning</i>	0.060	0.060	0.086*	0.059	0.081*	0.069	0.014	0.101*
<i>Socialization</i>	0.003	-0.006	-0.013	0.001	-0.000	-0.003	-0.067	0.037
<i>Transportation</i>	0.017	-0.005	0.020	-0.019	0.023	0.011	-0.062	0.056

Table 2: Effects of Each Property on Specified Perceptions. Each column represents a model, with the effect as the dependent variable and properties as independent variables. Using 'Comprehensibility' as the baseline property for comparison, the values indicate how the presence of each property changes perception relative to this baseline. Significance levels: * $p < .05$, ** $p < .01$, * $p < .001$**

straightforward wording used for other properties, the term "anthropomorphism" may have appeared peculiar and overly emphasized the human entity to participants. Future research should account for these potential confounding factors to better understand the impact of anthropomorphic design in XAI.

Although previous literature has favored *contrastivity* [42, 58, 62], our findings indicate that it has an overall negative effect. Such effect may be attributed to the cognitive load that contrastivity imposes on users. In everyday scenarios, users have limited cognitive resources, and the additional demands of contrastivity can diminish their overall experience (**satisfaction** and **importance**) and hinder their ability to perform routine tasks effortlessly (**effectiveness** and **efficiency**).

Comparing the results for **effectiveness** and **efficiency**, we find that confidence, translucence, contrastivity, and anthropomorphism strongly negatively contribute to both perceived effectiveness and efficiency. These XAI features should be avoided to deliver an effective and efficient user experience. Interestingly, *compactness* is negatively correlated with perceived **effectiveness**, and *completeness* is negatively correlated with perceived **efficiency**. In other words, we should steer clear of overly compact wording to deliver an effective explanation and refrain from providing exhaustive explanations to deliver an efficient one.

5.2 Reverse Engineering XAI with User-Centered Goals

One central challenge in XAI design is identifying an optimal balance within a multidimensional design space that addresses trade-offs between various design objectives for diverse user groups. Both prior research [3, 6, 43, 52] and our study suggests that no single design can simultaneously satisfy all these varied demands.

To address this, we propose the concept of goal-oriented XAI engineering (Fig. 2). Instead of selecting criteria arbitrarily based on

intuition or anecdotal evidence (Fig. 1), goal-oriented XAI engineering begins by defining the underlying values of XAI development by explicating the specific user effects that XAI aims to achieve. Based on these defined goals, a subset of XAI criteria is selected to guide the technical optimization and evaluation of XAI systems, tailored to specific contexts and user groups.

Practically, our study provides a summary in Table 8 of Appendix B.2, serving as a starting point for implementing goal-oriented XAI engineering in everyday scenarios. For example, to develop a scrutable XAI system that allows users to correct the system's outputs, prioritizing translucency and coherence is essential, whereas compactness and anthropomorphism should be de-prioritized. Future research could empirically examine these recommendations through field studies. Additionally, further investigation into the correlations among different goals is necessary. For example, optimizing for scrutability may inadvertently make a system more persuasive and potentially manipulative, or less satisfactory to users. Understanding these interactions will inform how design can be improved to balance multiple goals.

6 Limitation and future direction

The contribution of our study lies in soliciting people's opinions to illustrate the effects of XAI design criteria. While self-reported opinions provide valuable insights and enable exploration of a broad problem space, they may lack ecological validity in real-world application settings, as argued in previous work [40]. Nevertheless, our statistical methods were conducted rigorously and demonstrated good reliability and validity, serving as an important initial step toward scoping down such a vast problem space. Future work should consider complementary approaches, such as experimental studies or longitudinal research, to further systematically formalize the impact of XAI design criteria within specific contexts and over extended periods.

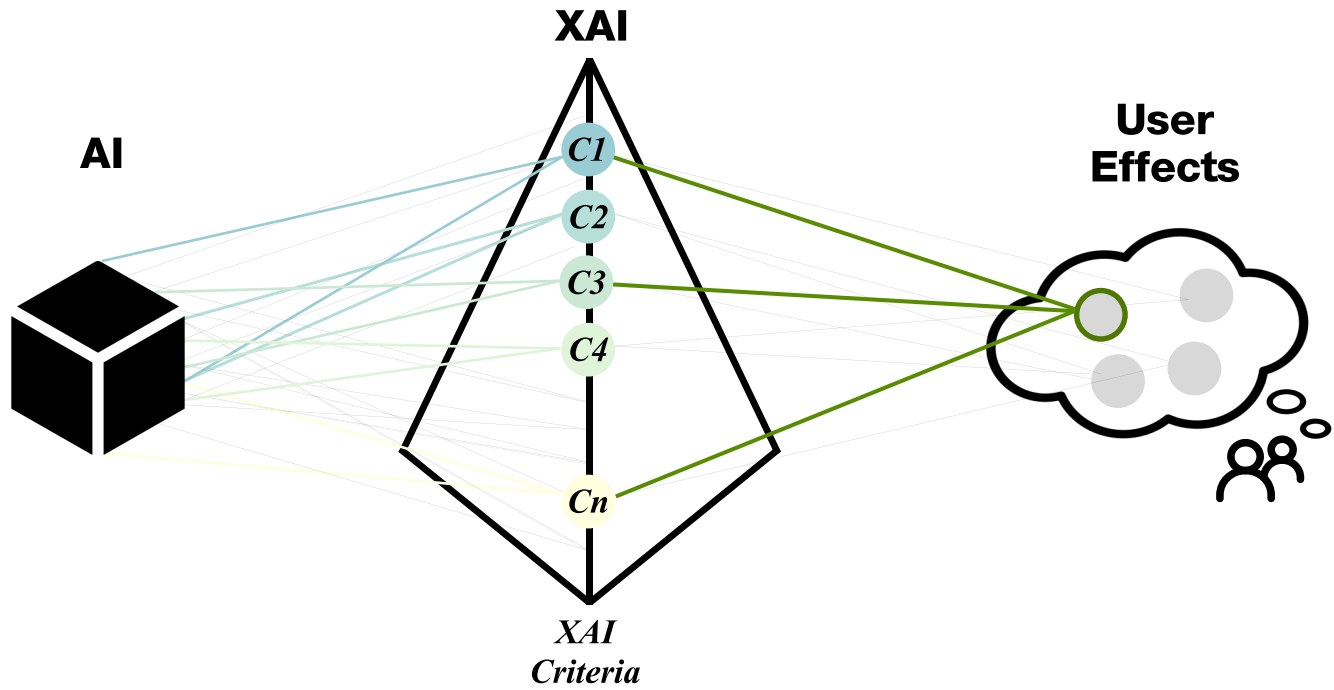


Figure 2: Goal-driven XAI Engineering, which begins by explicating the specific user effects that XAI aims to achieve. Based on these defined goals, a subset of XAI criteria (e.g., C1, C3) is selected to guide the technical optimization and evaluation of XAI systems (e.g., what kind of information about black-box AI models should be presented, and how should it be presented).

7 Conclusion

This study uncovers the effects of various XAI design properties on end-users' perceptions, addressing the disconnect between the techno-centric XAI requirements commonly discussed in the literature and their human-centered impacts. By offering empirical evidence on user effects in everyday scenarios for the general public rather than in high-stake conditions for experts—we emphasize comprehensibility and coherence as the most desired and fundamental properties of XAI design. In contrast, the frequently advocated properties such as contrastivity and anthropomorphism might have overall negative effects on user experience. These results also underscore the vital importance of aligning XAI design with users' goals and expectations. Based on these empirical findings, this research advocates for goal-oriented XAI engineering, providing actionable guidance on prioritizing and balancing design criteria to support specific user goals, which will ultimately promote a more meaningful integration of XAI into end-users' daily life.

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A Appendix: Study Design

A.1 Daily Scenarios

To represent XAI applications that impact the general public on a more day-to-day basis, we selected six common daily scenarios where AI is either already deeply integrated (e.g. recommendation systems [22, 53]) or has been widely discussed by researchers for potential integration (e.g. car automation [34]), including socialization, entertainment, healthcare, finance, learning, and transportation (Table 3).

A.2 XAI Properties

Table. 4, along with XAI literature that discussed each property.

B Appendix: Statistical Reports

B.1 Demographics Information

Table 5 shows participants' demographic information breakdown, including gender, race/ethnicity, education, employment, and age.

B.2 Mean values of properties

Table 8 presents the mean ratings of each perception by user participants for each property. Properties such as comprehensibility, translucence, and coherence achieve higher rankings across multiple perceptions, highlighting their consistently important role in enhancing positive user effects. In contrast, anthropomorphism consistently ranks the lowest in six out of eight perceptions, indicating that this property might detract from user perceptions in most contexts.

B.3 Tests for sample size

Table 7 illustrates each model's intraclass correlation coefficients (ICC), conditional R^2 values, and power simulations using 100 iterations. These validated the dataset's adequacy for detecting meaningful effects, ensuring robust and reliable results.

B.4 Effects of properties on each perception

Fig. 3, Fig. 4, Fig. 5, and Fig. 6 demonstrate the property coefficients for each perception with 95% confidence intervals. Their x-axis represents estimated coefficients, and the y-axis lists the property variables.

Scenario	Description used in the study	Literature Ex-amples
Socializing	AI has been integrated into human socialization, like recommending friends on social media, connecting potential romantic partners on dating apps, matching players in online games, participating in household family activities, etc.	[22, 36]
Entertainment	In the entertainment industry, AI has been employed to recommend personalized content such as movies and music. It also generates adaptive, immersive experiences tailored to individual choices, as seen in interactive books and games.	[32]
Healthcare	AI applications in healthcare include health status monitoring, diagnostics, personalized treatment, etc.. AI algorithms can analyze medical data to aid in diagnoses, predict patient outcomes, and tailor treatment plans to individual profiles.	[53]
Finance	AI is widely utilized in finance for managing personal finances, providing advice on budgeting, saving, and investing. It also powers automated 24/7 customer service and sends customized notifications and alerts. Additionally, AI is used to assess credit scores, automate loan approvals, and tailor insurance policies.	[40]
Learning	AI offers a wealth of applications that enhance educational experiences. It provides personalized learning paths, automates grading and feedback, and supports virtual one-on-one tutoring and exercises. Additionally, AI boosts engagement through gamification, making learning more interactive and enjoyable.	[8, 24, 31]
Transportation	AI has been widely utilized in the transportation sector, enabling autonomous driving, advanced cruise control, lane keeping, and automatic parking. In public transportation, AI optimizes transit routes and schedules, and provides users with real-time updates on service statuses. Additionally, ride-sharing and hailing services leverage AI for dynamic pricing and efficient ride matching.	[34]

Table 3: Scenarios and Descriptions used in the survey study, with corresponding examples from related literature.

Design Property	Definition	Literature
Comprehensibility	The explanation is presented in a clear and easy-to-understand manner.	[12, 40] Composition [43] Clarity [48] Understandability [27] Explicitness [1]
Coherence	The explanation should be consistent with the evidence and with itself.	[30, 37, 40, 43, 52]
Soundness	The explanation accurately reflects how the system works.	[35] Correctness [43] Faithfulness [1, 40] Fidelity [12, 48]
Completeness	The explanation fully covers the AI model.	Representative [12, 40, 43]
Translucence	The explanation is transparent about its limitations.	[12, 40] Contextfulness [52]
Confidence	The explanation reflects certainty, accuracy, or other probability information of AI's behaviors.	[43] (Un)certainity[40]
Compactness	The explanation provides only necessary information.	[43] Parsimony [40]
Contrastivity	The explanation should clearly show the difference when compared to other events or targets.	[43, 50, 62]
Anthropomorphism	The explanation is verbally human-like.	[14, 42]

Table 4: Design Properties (synonyms were provided if used in the referenced literature)

	Number of participants	Percentage (%)
Gender		
Woman	43	49.42
Man	37	42.52
Non-binary	4	4.60
Trans	3	3.45
Race/Ethnicity		
African	1	1.15
African-American or Black	9	10.34
Asian-American	6	6.90
East Asian	3	3.45
Hispanic or Latino/a/x	9	10.34
Indigenous American or First Nations	1	1.15
Middle Eastern	1	1.15
Not listed	2	2.30
South Asian	1	1.15
Southeast Asian	3	3.45
White	51	58.62
Education		
High school graduate	8	9.20
Associate degree in college	5	5.74
Bachelor's Degree in college	38	43.68
Master's Degree	8	9.20
Doctoral degree	1	1.15
Some college	23	26.44
Professional degree	1	1.15
Technical, vocational, or trade school	3	3.45
Employment Status		
Employed Full-time	34	39.08
Employed Part-time	19	21.84
Looking for work	13	14.94
Not in the paid workforce	16	18.39
Others	15	17.24

Table 5: Summary of demographic profiles

Min	1st Quartile	Median	Mean	3rd Quartile	Max
18	26	32	32.7	39.5	63

Table 6: Age Distribution (years)

	Scrutable	Transparent	Trustful	Persuasive	Effective	Efficient	Satisfying	Important
Conditional R²	0.387	0.383	0.403	0.377	0.340	0.341	0.364	0.291
Adjusted ICC	0.329	0.275	0.349	0.356	0.312	0.309	0.345	0.126

Table 7: Conditional R² and Adjusted ICC for each perception.

	Scrutable	Transparent	Trustful	Persuasive
1	Translucence (0.99)	Translucence (1.47)	Translucence (1.22)	Confidence (1.02)
2	Coherence (0.96)	Soundness (1.28)	Soundness (1.22)	Coherence (1.00)
3	Soundness (0.95)	Completeness (1.19)	Coherence (1.19)	Comprehensibility (0.99)
4	Completeness (0.95)	Coherence (1.06)	Completeness (1.10)	Soundness (0.91)
5	Comprehensibility (0.77)	Comprehensibility (1.03)	Comprehensibility (1.07)	Completeness (0.90)
6	Constrastivity (0.68)	Constrastivity (0.83)	Confidence (1.00)	Constrastivity (0.81)
7	Confidence (0.54)	Confidence (0.74)	Constrastivity (0.80)	Translucence (0.80)
8	Compactness (0.18)	Compactness (0.32)	Anthropomorphism (0.56)	Anthropomorphism (0.69)
9	Anthropomorphism (0.14)	Anthropomorphism (0.27)	Compactness (0.47)	Compactness (0.49)

	Effective	Efficient	Satisfying	Important
1	Comprehensibility (1.30)	Comprehensibility (1.25)	Comprehensibility (1.14)	Comprehensibility (1.63)
2	Coherence (1.23)	Compactness (1.25)	Coherence (1.02)	Coherence (1.52)
3	Completeness (1.13)	Coherence (1.11)	Completeness (0.93)	Soundness (1.24)
4	Soundness (1.12)	Soundness (0.98)	Soundness (0.90)	Completeness (1.23)
5	Translucence (0.96)	Completeness (0.86)	Translucence (0.74)	Translucence (1.17)
6	Confidence (1.02)	Confidence (0.85)	Confidence (0.85)	Confidence (1.14)
7	Compactness (0.94)	Translucence (0.86)	Compactness (0.82)	Compactness (0.73)
8	Constrastivity (0.85)	Constrastivity (0.68)	Constrastivity (0.60)	Constrastivity (0.57)
9	Anthropomorphism (0.76)	Anthropomorphism (0.69)	Anthropomorphism (0.78)	Anthropomorphism (0.27)

Table 8: Mean values of properties by perceptions with rankings. The top two properties are shaded green and the bottom ones red.

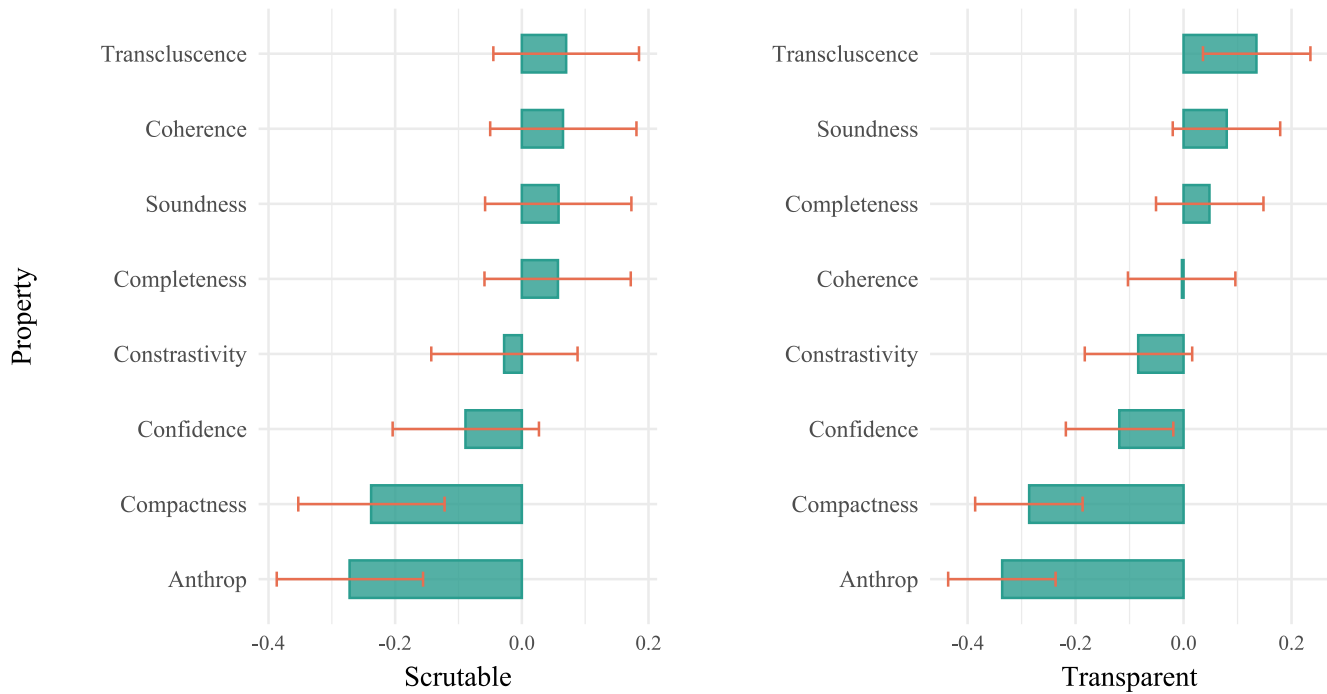


Figure 3: Property coefficients for "Scrutable" and "Transparent" with 95% confidence intervals. The x-axis represents estimated coefficients, and the y-axis lists the property variables.

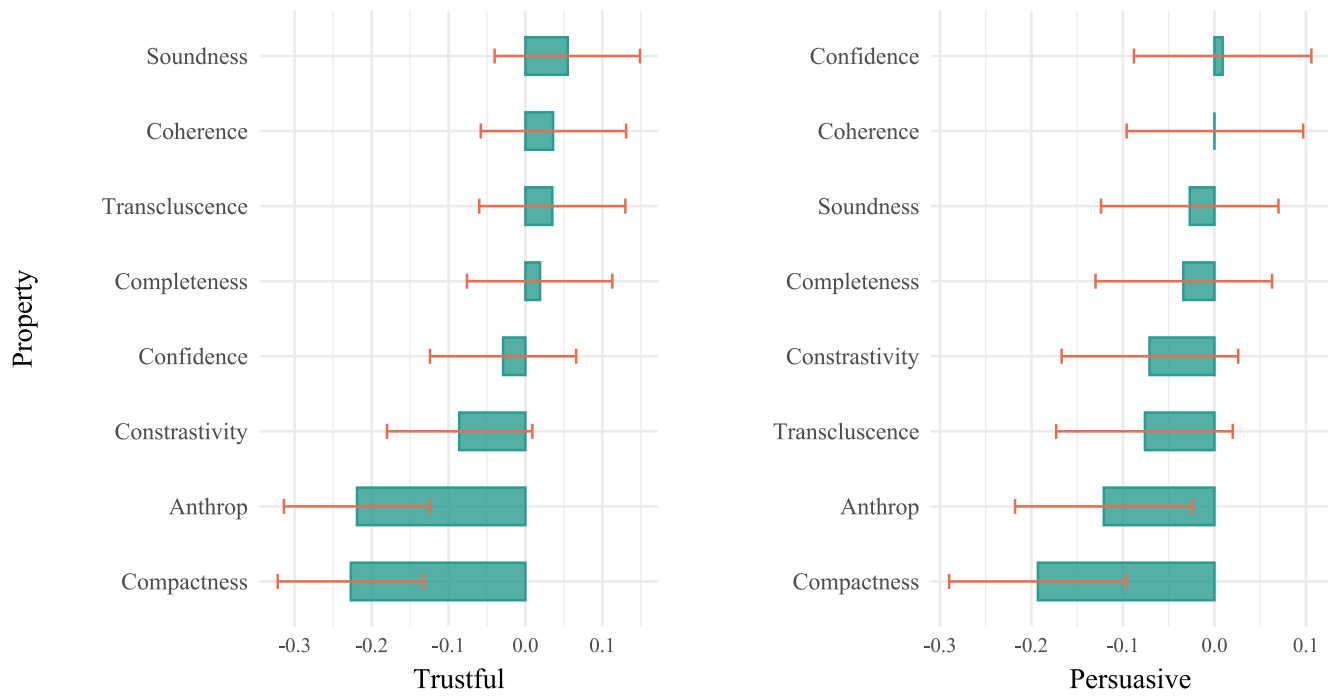


Figure 4: Property coefficients for "Trustful" and "Persuasive" with 95% confidence intervals

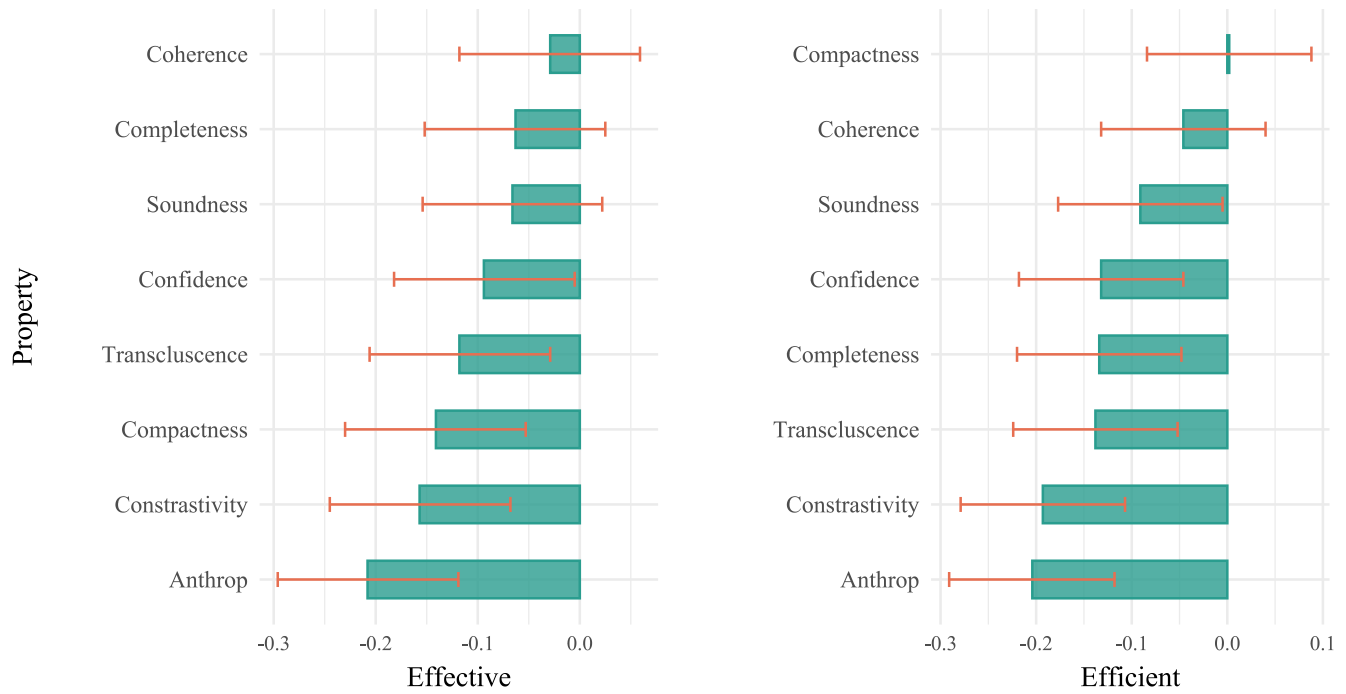


Figure 5: Property coefficients for "Effective" and "Efficient" with 95% confidence intervals

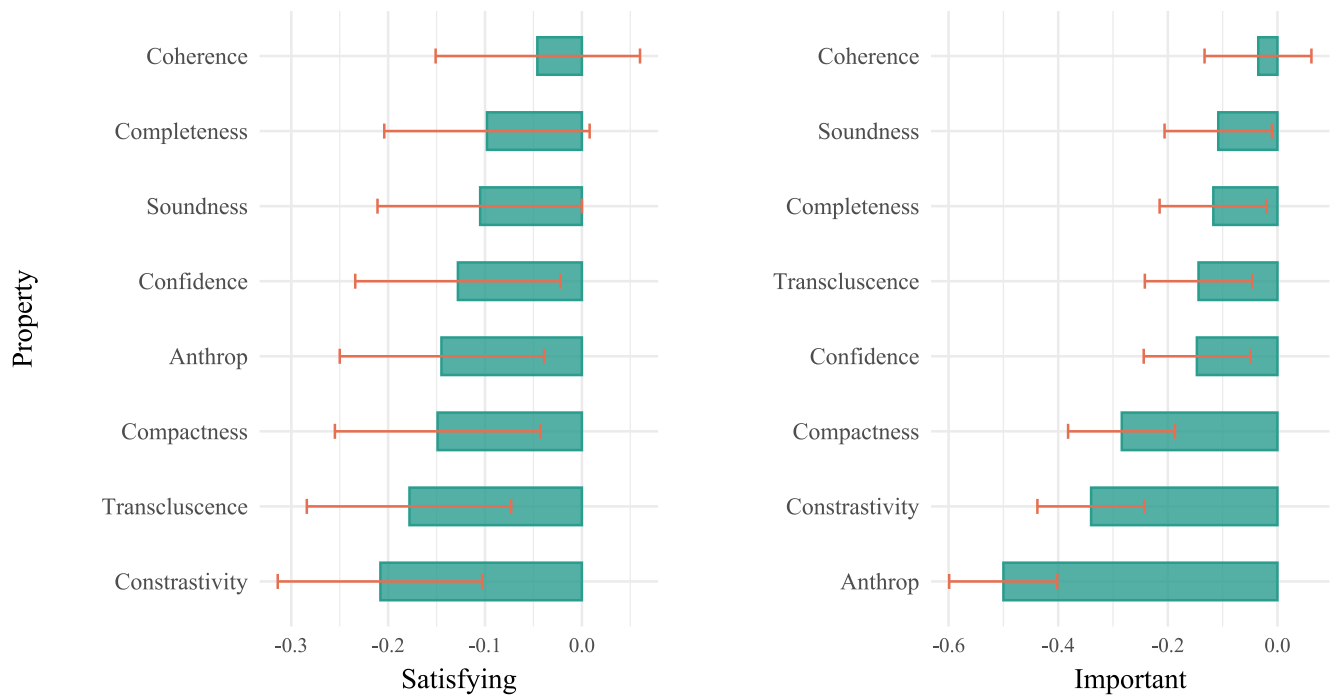


Figure 6: Property coefficients for "Satisfying" and "Important" with 95% confidence intervals